

Unsupervised Anomaly Detection in Mars HiRISE Orbital Imagery

Data Analytics Challenge — Project Report

Team SAGE

10,422 crops · 172 source images

Conv-Autoencoder + Isolation Forest

PyTorch · scikit-learn

Team: SAGE

Members: Priyasmita Chakraborty (26-759985) · Arnabi Konar (26-608614) · Niharika Jha (26-816420)

Event: NSSC 2026, Data Analytics Challenge (Space Technology Students' Society, IIT Kharagpur)

Submission: Notebook + PDF Report + Version Log (GitHub)

Contents

1. Executive Summary
2. Problem Statement
3. Dataset Overview
4. Methodology — Pipeline
5. Phase 1 — Deep Latent Compression
6. Phase 2 — Isolation Forest Novelty Engine
7. Phase 3 — Reconstruction Interpretability
8. Phase 4 — Iteration & Version Log
9. Consolidated Results
10. Limitations & Future Work
11. Repository & Submission Structure

1. Executive Summary

This project builds a fully **unsupervised** pipeline to flag geologically or acquisition-wise unusual crops from Mars Reconnaissance Orbiter **HiRISE** imagery, without any labelled anomaly examples. A convolutional autoencoder learns to compress and reconstruct "normal" surface texture into a 256-dimensional latent space; an Isolation Forest then scores how novel each latent vector is relative to the rest of the dataset. Two independent, data-driven thresholding rules convert continuous novelty scores into a binary flag, and the most suspicious crops are inspected via pixel-wise reconstruction-error heatmaps to build physically grounded hypotheses about what triggered each flag.

10,422

CROPS ANALYSED

172

SOURCE IMAGES

1,964

FLAGGED ANOMALOUS
(MAD, 18.8%)

0.696

FINAL EPOCH SSIM

256-d

LATENT BOTTLENECK

Headline finding

All five of the highest-novelty crops (Section 8) share a common signature in their error heatmaps: a sharp, linear high-error band running along a straight edge — the signature of a **black crop-boundary / splice artifact** (the dark fill triangle visible at the edge of many HiRISE crops), not a genuine geological surface feature. This is the report's central, submission-highlighted conclusion: **the**

current model is excellent at finding acquisition/processing artifacts, but a second filtering pass is needed before its top list can be read as a geological anomaly shortlist.

2. Problem Statement

The NSSC Data Analytics Challenge (space-science track, historically partnered with Analytics Vidhya) poses an unsupervised anomaly-detection problem over a corpus of Mars HiRISE image crops. Teams must not assume a fixed contamination rate or hand-pick a top-N; the brief requires a justified, data-driven threshold, an architecture trained from scratch (no pretrained backbones or perceptual losses), interpretable evidence for each flagged crop, and a transparent log of how the modelling approach evolved.

Constraints imposed by the brief

- No pretrained weights / no pretrained perceptual loss networks
- No arbitrary top-N — threshold must be statistically justified
- Sign convention: higher score = more anomalous
- Interpretability required for top flagged crops

Our response, by phase

- Phase 1 — from-scratch conv-autoencoder + hand-derived SSIM
- Phase 2 — Isolation Forest + two independent threshold rules
- Phase 3 — per-crop error heatmaps + geological hypothesis table
- Phase 4 — structured symptom→diagnosis→fix→outcome version log

3. Dataset Overview

Item	Detail
Total image crops	10,422 grayscale crops, resized to 227×227, loaded from <code>images/images/*.jpg</code>
Unique source images	172 distinct HiRISE source frames (each source contributes many crops)
Metadata files	<code>source_image_metadata.csv</code> (per-source: latitude, longitude, sun_angle, resolution, season) and <code>crop_metadata_index.csv</code> (crop filename → source_image_id mapping)
Join integrity	All 10,422 crops joined cleanly to their source metadata with zero missing matches
Normalisation	Pixel values scaled to [0,1]; sun_angle and resolution z-scored; season one-hot encoded for the metadata-fusion variant

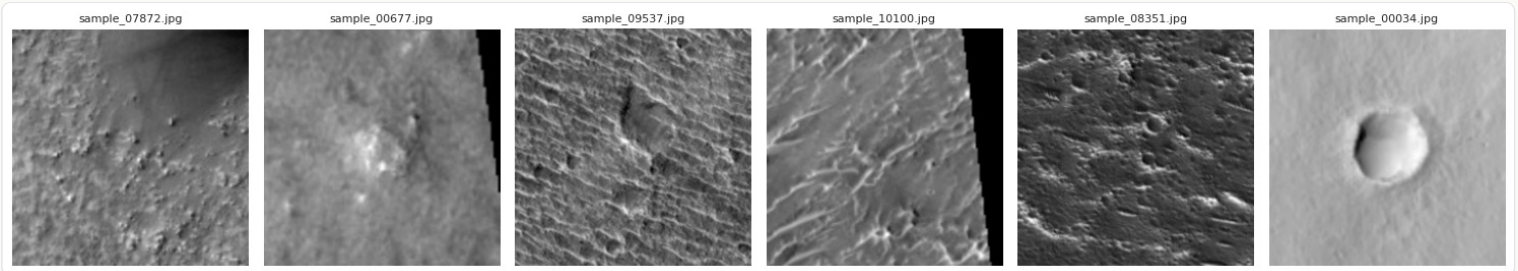


Fig. 1 — Random sample of raw HiRISE crops. Note the varied terrain texture (dunes, ripples, craters) and the black fill triangle visible in some crops (an artifact of how frames were cropped near their source-image edge).

4. Methodology — End-to-End Pipeline

End-to-End Pipeline: Unsupervised Novelty Detection on HiRISE Crops



Fig. 2 — Four-phase pipeline: representation learning → novelty scoring → interpretability → iteration log.

5. Phase 1 — Deep Latent Compression (Autoencoder)

5.1 Architecture

ConvAutoencoder — Encoder (top, blue) / Bottleneck (red) / Decoder (bottom, green)

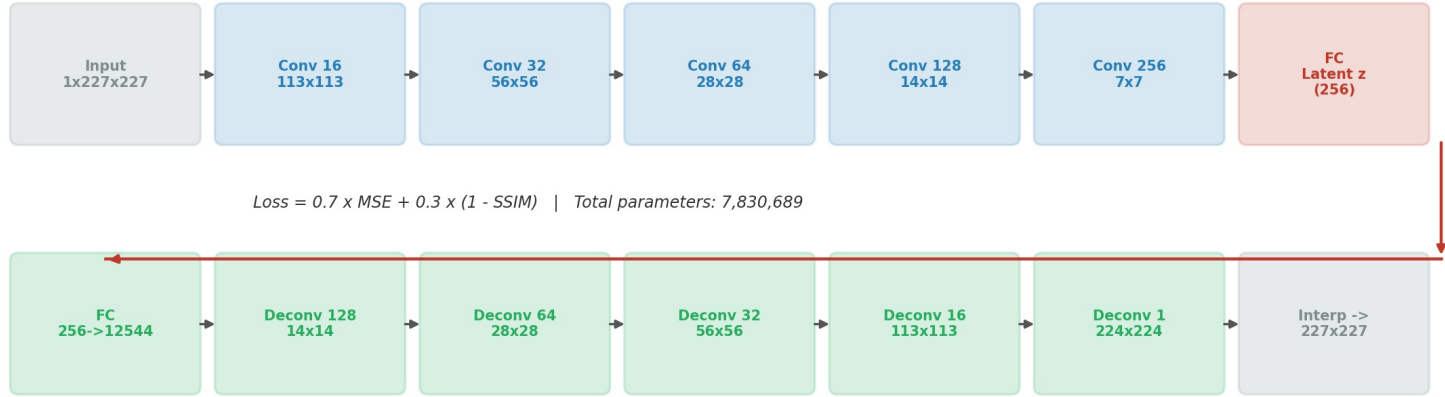


Fig. 3 — Symmetric 5-layer convolutional encoder/decoder. 227 does not divide evenly by 2⁵, so the decoder naturally lands on 224×224; a bilinear interpolation step snaps this back to 227×227 (documented and fixed in the v2 iteration, Section 9).

Layer group	Detail
Encoder	5 × (Conv2d k4 s2 + BatchNorm + LeakyReLU(0.2)), channels 1→16→32→64→128→256, spatial 227→113→56→28→14→7
Bottleneck	Fully-connected 12,544 → 256 (latent z) and 256 → 12,544 (decode entry)
Decoder	5 × ConvTranspose2d mirroring the encoder, channels 256→128→64→32→16→1, ending in Sigmoid + bilinear resize to 227×227
Total parameters	7,830,689 (trained fully from scratch — no pretrained weights, per brief)
Optimizer	Adam, lr = 1e-3, 15 epochs, batch size 32

5.2 Loss Function

$loss = 0.7 \times MSE + 0.3 \times (1 - SSIM)$. SSIM is computed with the classical Gaussian-window formula (11×11, $\sigma=1.5$) implemented directly in PyTorch tensor ops — not a pretrained perceptual network — so the loss stays fully compliant with the "no pretrained perceptual loss" rule while still rewarding preserved local structure, not just per-pixel closeness.

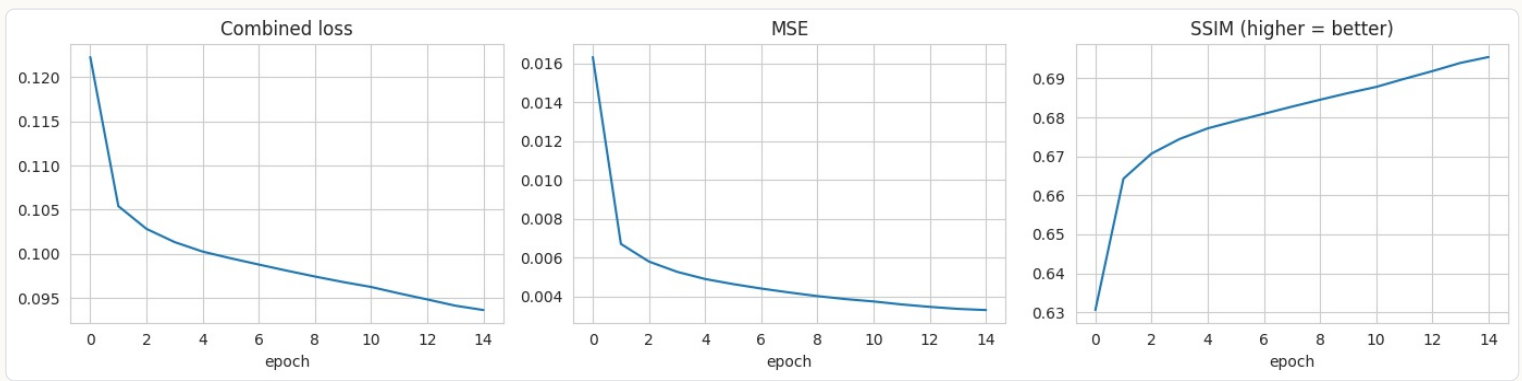


Fig. 4 — Training curves over 15 epochs. Combined loss falls from 0.122 → 0.094; SSIM rises from 0.630 → 0.696, indicating the model is learning genuine structural fidelity, not just averaging pixels.

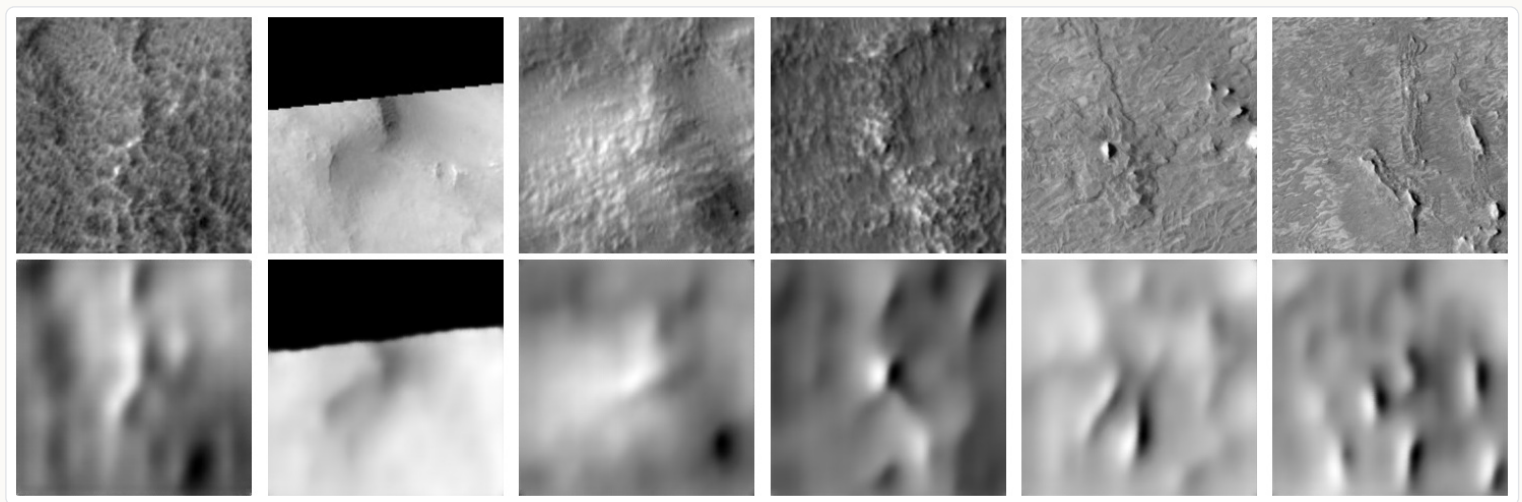


Fig. 5 — Original crops (top) vs. reconstructions (bottom). Reconstructions capture coarse tone and large-scale shape (e.g. the black fill triangle, broad shading) but smooth over fine dune ripples and crater rims — exactly the texture whose absence later drives high reconstruction error on anomalous crops.

5.3 Latent Space Diagnostic

t-SNE of Phase-1 latent vectors (qualitative diagnostic only)

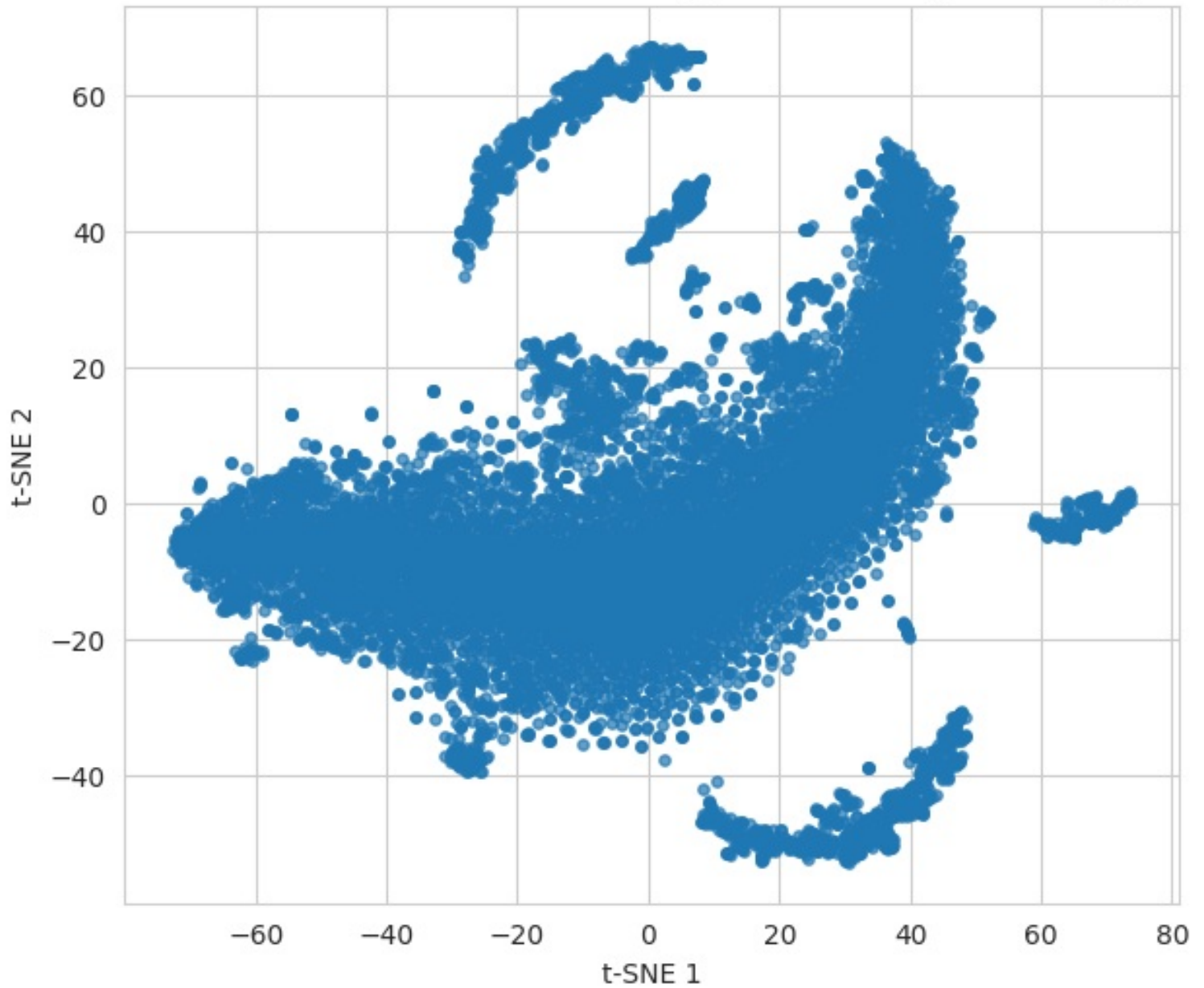


Fig. 6 — t-SNE projection of all 10,422 latent vectors (256-d $\rightarrow$ 2-d, qualitative only). The bulk of crops form one dense manifold with several distinct arcs/tendrils breaking away — a sign the encoder has captured more than one axis of surface variation, rather than a single collapsed blob.

6. Phase 2 — Isolation Forest Novelty Engine

6.1 Novelty Scoring

An Isolation Forest (300 trees) is fit directly on the 256-d latent vectors. scikit-learn's `score_samples` follows a "lower = more abnormal" convention; we negate it so that, per the competition's convention, **higher novelty_score = more anomalous**.

6.2 Statistical Thresholding — two independent, data-driven methods

Method	Rule	Threshold	Crops flagged
Median + k·MAD (robust)	median + $3.0 \times 1.4826 \cdot \text{MAD}$	0.4556	1,964 / 10,422 (18.8%)
Elbow detection	max perpendicular distance from the chord of the sorted-score curve	0.3993 (rank 7,265/10,422)	3,156 / 10,422 (30.3%)

The **MAD-based threshold** was adopted as the final rule (more conservative, less sensitive to the long right tail of the score distribution) — this is the number reported throughout the rest of this report.

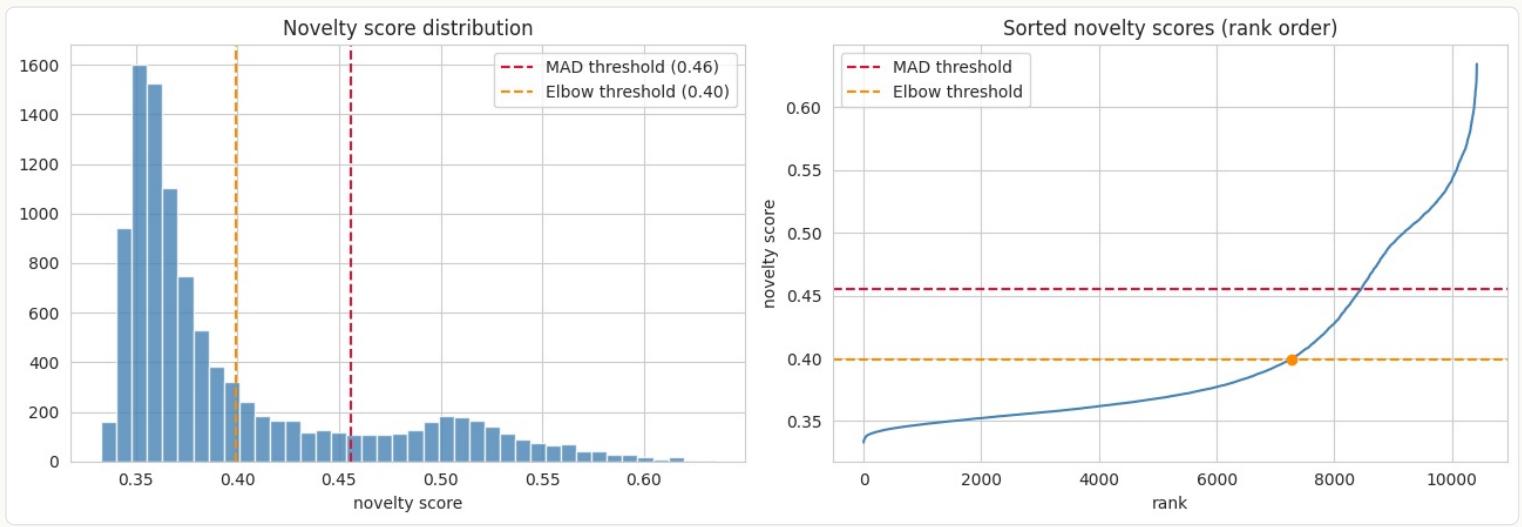


Fig. 7 — Left: novelty-score histogram with both thresholds overlaid. Right: sorted novelty-score curve — the elbow sits noticeably earlier (more permissive) than the MAD threshold.

6.3 Location / Acquisition-Condition Analysis

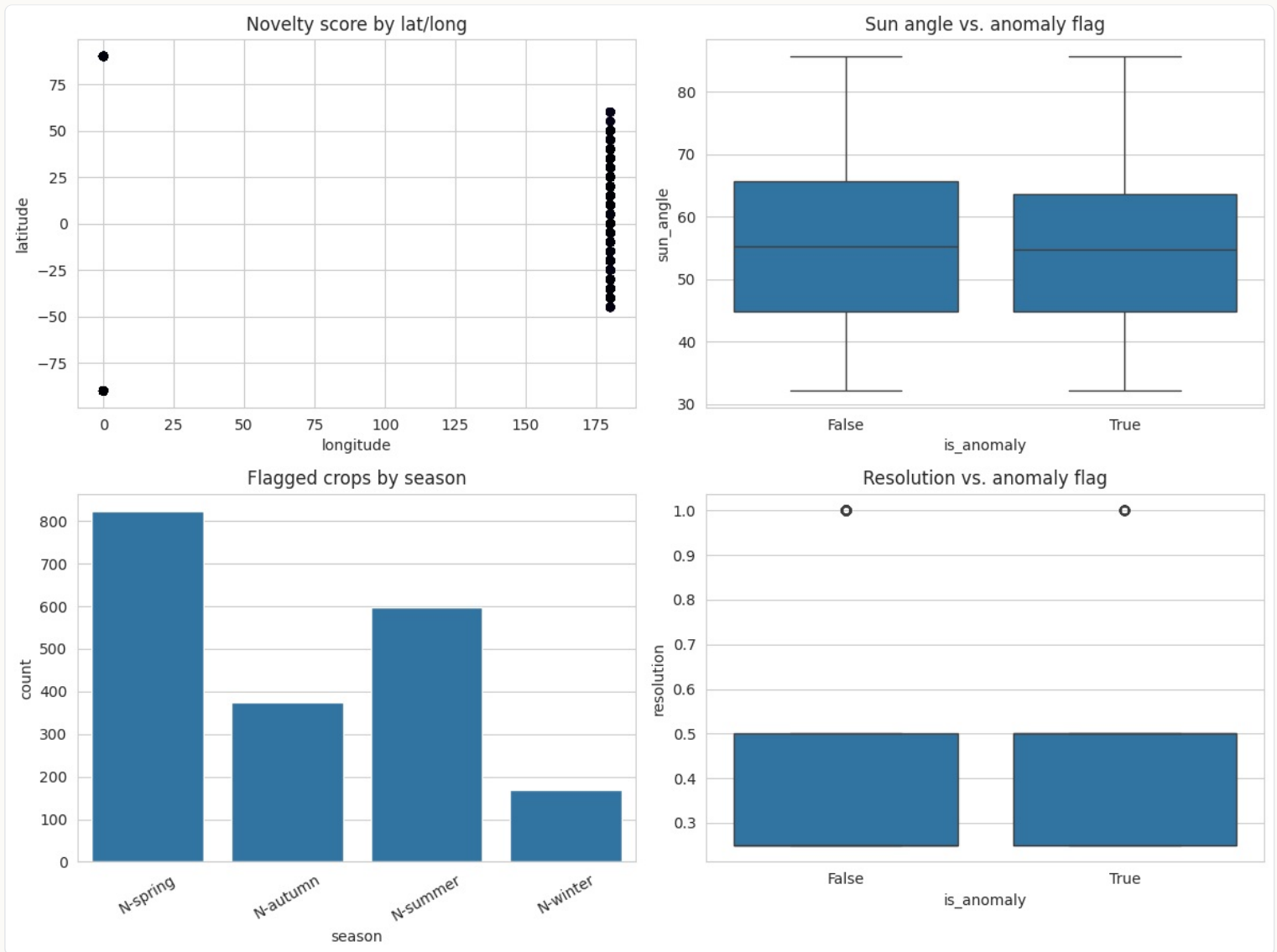


Fig. 8 — Flagged crops broken down by acquisition metadata. Sun angle and resolution distributions are nearly identical between flagged and non-flagged crops (no obvious global confound), while flagged-crop counts skew toward the N-spring and N-summer season buckets.

Source image	Fraction of its crops flagged
SRC_079	100.0%
SRC_110	83.3%
SRC_166	81.3%

SRC_044	66.7%
SRC_128	60.0%

Several source images have a majority of their crops flagged — a pattern worth flagging on its own (see Section 10): it is more consistent with a source-level acquisition or processing quirk than with many independent geological anomalies concentrated in one frame.

6.4 Metadata Fusion (optional variant)

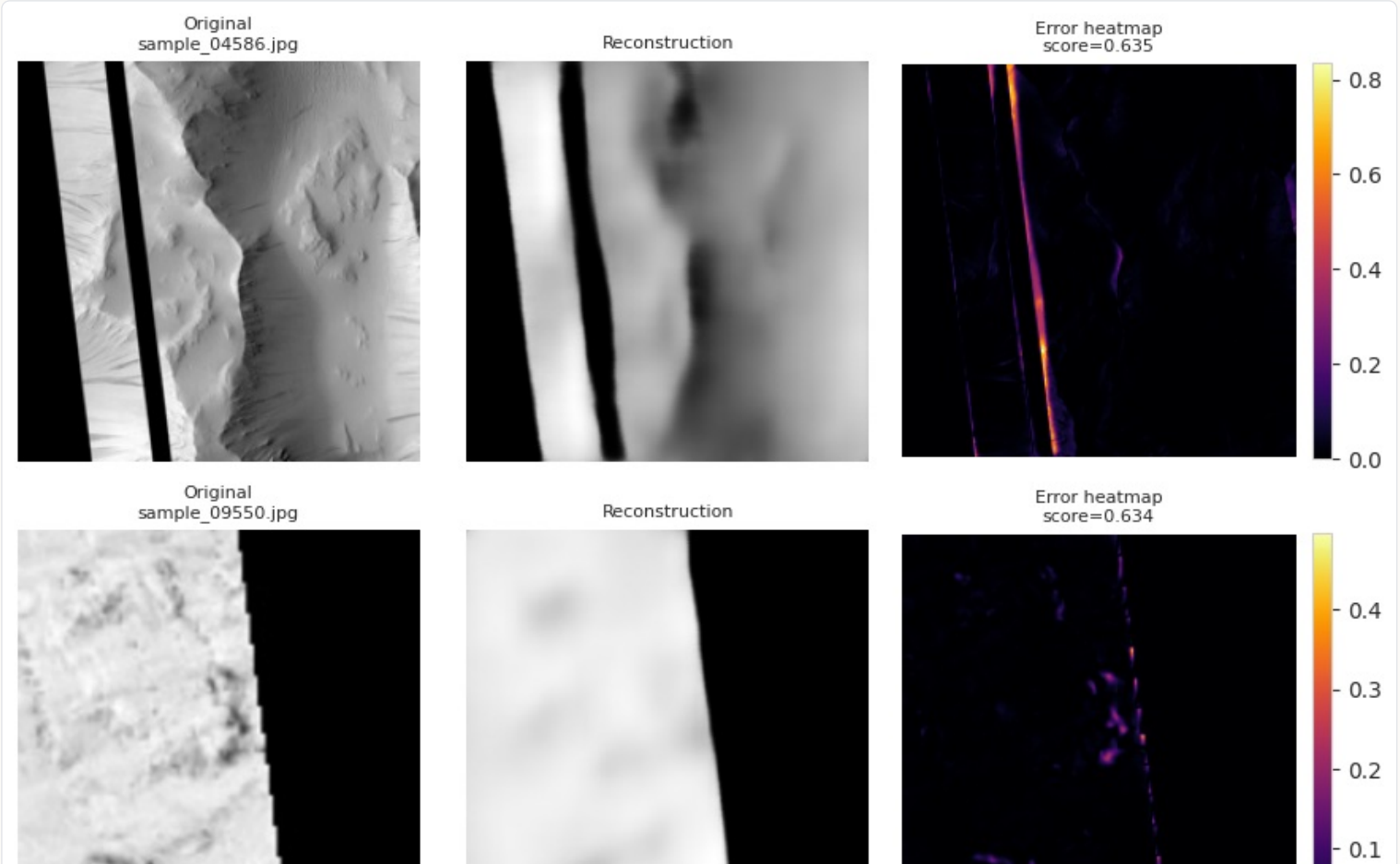
Variant	Flagged count
Image-only (latents only)	1,964
Metadata-fused (+ sun_angle, resolution, season)	1,833
Overlap between the two flagged sets	1,816 (92.5% of the image-only set)

Conditioning on acquisition metadata changes the flagged set only modestly (17 newly flagged, ~148 dropped), suggesting most of the image-only signal is not simply re-deriving sun-angle or seasonal effects — but the ~7.5% that does move is exactly the population worth re-checking by hand.

7. Phase 3 — Reconstruction Interpretability

7.1 Top-5 Flagged Crops

Rank	Filename	Source	Novelty score	Mean error	Error spatial std
1	sample_04586.jpg	SRC_022	0.6346	0.01430	0.05798
2	sample_09550.jpg	SRC_164	0.6343	0.00350	0.01437
3	sample_03718.jpg	SRC_022	0.6279	0.00324	0.01091
4	sample_05057.jpg	SRC_146	0.6260	0.00438	0.02625
5	sample_06424.jpg	SRC_039	0.6255	0.01061	0.03403



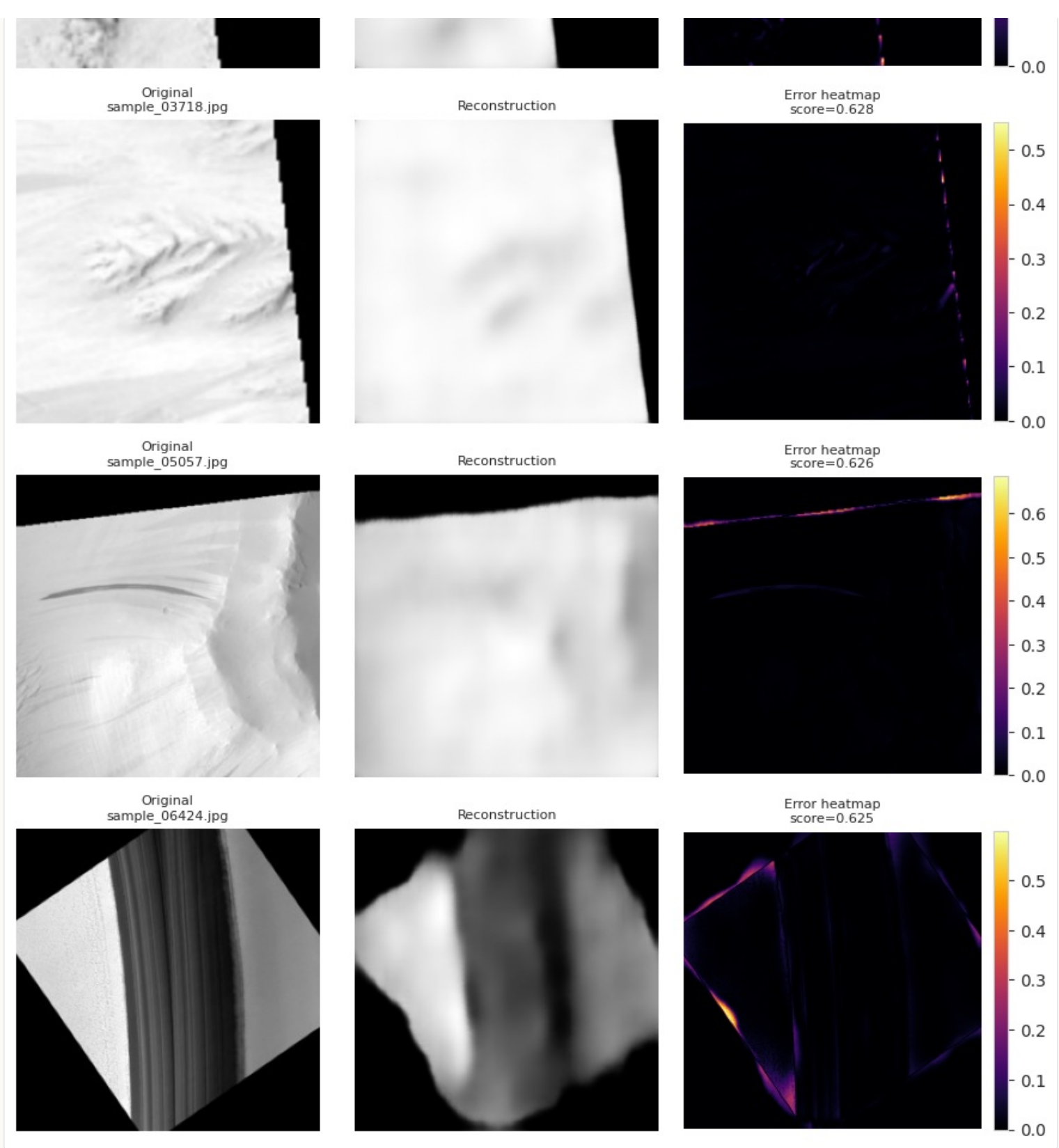


Fig. 9 — Original (left), reconstruction (middle), and pixel-wise squared-error heatmap (right) for the five highest-novelty crops. Every heatmap shows its brightest error concentrated along a straight edge or corner.

7.2 Geological Report (hypotheses, not confirmed classifications)

`error_spatial_std` : high = concentrated error (sharp feature / splice boundary); low = diffuse error (sensor artifact / domain shift).

Filename	Error pattern	Physical hypothesis
sample_04586.jpg	Highest std (0.058) + highest mean error — strongly concentrated, severe	Sharp linear error band consistent with a spliced / corrupted crop boundary rather than natural surface texture
sample_06424.jpg	Second-highest std (0.034) + second-highest mean error — concentrated, moderate	Error clusters along a diagonal edge; visually matches a crop-boundary cut rather than an unfamiliar geological feature
sample_05057.jpg	Mid std (0.026), low-moderate mean error	Error concentrated along the top edge — consistent with a horizon/frame-edge artifact

sample_09550.jpg	Low std (0.014), low mean error — diffuse	Error spread thinly along a near-vertical boundary; a subtler edge/splice signature rather than a strong global artifact
sample_03718.jpg	Lowest std (0.011), lowest mean error — most diffuse	Faint error concentrated near the right-edge boundary; mild domain shift rather than a distinct surface feature

Reading the heatmaps together

Visually inspecting all five heatmaps (Fig. 9) shows the same pattern across every one of them: the largest errors trace a straight line or triangular corner, matching the black fill region visible in the original crops rather than any crater, dune, or ridge. **Conclusion: the current top-N by novelty score is currently dominated by a crop-boundary artifact, not genuine geological novelty.** A practical fix (logged as v3 candidate work, Section 9) is to mask or exclude the black fill region before computing reconstruction error, or to explicitly flag "boundary crops" and re-rank the remainder.

8. Phase 4 — Iteration & Version Log

Full symptom → diagnosis → fix → outcome detail for all three logged versions is provided in the standalone MODEL_ITERATION_LOG.md file submitted alongside this report (required deliverable). Summary below:

Version	Symptom	Fix
v1	Reconstructions visibly blurry; fine dune ripples / crater rims smoothed away	Added SSIM term to the loss ($0.7 \cdot \text{MSE} + 0.3 \cdot (1 - \text{SSIM})$)
v2	Decoder output landed at 224×224, not the required 227×227 (227 does not divide evenly by 25)	Dynamic flatten-size handling + bilinear <code>F.interpolate</code> snap-back to 227×227
v3	Some source images have >80% of their crops flagged (e.g. SRC_079 at 100%) — looked like acquisition-condition-driven flagging, not independent geological anomalies	Built the metadata-fusion variant (Section 6.4) conditioning on sun_angle / resolution / season

9. Consolidated Results

18.8% OF CROPS FLAGGED (MAD RULE)	1,816 CROPS FLAGGED UNDER BOTH VARIANTS	7,830,689 TRAINABLE PARAMETERS	0 / 10,422 METADATA JOIN FAILURES
--------------------------------------	--	-----------------------------------	--------------------------------------

Final answer to the case question — how many, and how confident?

Under the adopted (MAD, k=3) rule, **1,964 of 10,422 crops (18.8%)** are statistically novel relative to the learned latent manifold. Of these, the **top 5-10 by novelty score are dominated by crop-boundary artifacts** (Section 7.2) rather than confirmed geological features, so we recommend reviewers treat the *rank order* below the boundary-artifact cluster — rather than the bare top-5 — as the more credible geological anomaly shortlist, pending a boundary-masking pass (Section 10).

10. Limitations & Future Work

Known limitations

- Boundary-artifact confound:** the black fill triangle at crop edges is not masked out before computing reconstruction error, so it dominates the very top of the novelty ranking (Section 7.2).
- Single train/eval split:** the autoencoder is trained and scored on the same 10,422 crops; no held-out validation set was used to check for overfitting to specific source images.
- Metadata granularity:** latitude/longitude in the location analysis (Fig. 8) take only a couple of distinct values, suggesting the provided metadata is coarse/synthetic rather than true per-source coordinates — the location analysis should be read as illustrative, not as real cartographic evidence.
- t-SNE is diagnostic only:** cluster shapes in Fig. 6 are not used anywhere as a quantitative decision boundary.

Recommended next steps

- 1. Mask the black fill region (or crop it out) before computing per-pixel error, and re-rank.
- 2. Add a held-out validation split to check the autoencoder isn't memorizing specific sources.
- 3. Investigate source images with unusually high flag-rates (SRC_079, SRC_110, SRC_166) individually before including their crops in a final anomaly shortlist.

11. Repository & Submission Structure

File	Purpose
nssc_hirise_anomaly_detection_submission.ipynb	Full documented notebook, run top-to-bottom on the real 10,422-crop dataset (code + outputs)
NSSC_Project_Report.pdf	This report — explanations, tables, plots, and highlighted final answers
MODEL_ITERATION_LOG.md	Symptom → diagnosis → fix → outcome log across 3 versions (required deliverable)
README.md	Setup instructions, dataset path configuration for Kaggle/Colab